

## Quicker Robotic Eyes

One of the reasons that we do not have robotic helpers in our homes already is because of how slow robotic eyes are. In an assembly line, the robots can work quickly because the environment does not change constantly. In unstructured environments, such as homes and hospitals, the robots can't adapt quick enough to ever-changing surroundings. Researchers from the University of Michigan have tackled this bottleneck, and come up with a solution an order of magnitude faster than previous approaches. This allows the robot to understand the position and orientation of objects within ten minutes. Previous robots would take about half an hour to do the same



thing. The new algorithm is called Pull Message Passing for Nonparametric Belief Propagation. Similar techniques

were already used in computer vision programs, but not robots because of how demanding they were in terms of processing power. The team managed to demonstrate the effectiveness of the algorithm, by effectively getting a robot to manipulate the shelves in a drawer, even when it was half covered by a blanket.

<https://digit.in/roboeyes>

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

## THIS MONTH IN SCIENCE:

FIND OUT ALL ABOUT THE MYSTERIOUS LIGHT PILLAR PHENOMENON, LEARN ABOUT THE ORIGINS OF GENE MAPPING, HOW MACHINE LEARNING IS USED TO DISCOVER NEW DRUGS AND WHAT IT TAKES TO VISUALISE A BLACK HOLE



## Autonomy can reduce traffic

Researchers have calculated the degree to which traffic can reduce if all the cars on the road are autonomous. The research shows that the traffic flow can improve by 35 per cent. If a human driver is introduced, the cars collectively move away to improve safety.

<https://digit.in/autotraf>

## Ancient toy inspires cutting edge tool

An ancient toy known as the whirligig, which is more than 5,000 years old, has inspired a cutting edge tool for scientists. The toy is a circular disc that spins because of twisted strings. Researchers at Georgia Tech found out that the simple toy, when modified, was very effective as a centrifuge. Along the disk, the researchers placed cavities, into which tubes containing samples could be placed. This



portable centrifuge, dubbed the 3D-Fuge could be manufactured at a fraction of the cost of other devices. While tabletop units cost between \$3,000 to \$5,000, the 3D-Fuge costs less than \$1! Four spe-

cies of plants and insects, previously unknown to science, were discovered through DNA samples that were isolated using the the 3D-Fuge by scientists in Peru. Gaurav Byagathvalli, co-inventor of the device said, "through this, we've realized how much of an impact simple yet effective tools can have, and hope this technology motivates others to continue along the same path."

<https://digit.in/3dfuge>



### Fabrics with chips

Researchers at MIT have embedded chips loaded with sensors into fabrics, which can be used for all sorts of composite products such as clothing, airplane wings or dressings for wounds. They can be used for a variety of purposes such as monitoring structural integrity, or the progress of a treatment.

<https://digit.in/fabware>



### Where do stars come from?

Astronomers studying a nearby galaxy, NGC 300 found that new stars are formed in clouds of hydrogen and dust, called molecular clouds. However, the new stars quickly disperse the clouds, in a relatively short span of time, in less than 1.5 million years.

<https://digit.in/strfrm>



### The colour of an extinct mouse

Thus far, it's been impossible to tell what colour a fossilised animal was. However, a team from Manchester University has done just that. Using high energy X-Rays, they investigated the traces of pigments in a fossilised mouse.

<https://digit.in/mousecolor>